

SonicBoom: 第三次アウトオブオーダプロ セッサ

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Introduction

- Deployment of high-performance superscalar, out-of-order (OOO) cores expanding into mobile and edge devices
- Security, power, performance, area need to be considered when evaluating new design
- An open-source hardware implementation of a superscalar, OOO core is invaluable

- Open source hardware implementation have numerous advantages over software models of high-performance cores
 - Demonstrate precise microarchitectural behaviors
 - Execute real applications for trillions of cycles
 - *Empirically* provide power and area measurements
 - Point of comparison for new microarchitectural platform

- Most of the current open-source hardware development framework only support simple in-order cores
 - Rocket
 - Ariane
 - Black Parrot
 - PicoRV32
- Modern mobile / server-class SOC is impossible without a full-featured, high performance implementation of a superscalar OOO core.

- Based on the data in table 1, SonicBOOM is the ***fastest* publicly available open-source** core

Table 1: Comparison of open-source out-of-order cores.

	Sonic BOOM	BOOMv2	riscy- OOO	RSD
ISA	RV64GC	RV64G	RV64G	RV32IM
DMIPS /MHz	3.93	1.91	?	2.04
SPEC06 IPC	0.86	0.42	0.48	N/A
Branch Pre-dictor	TAGE	GShare	Tourney	GShare
Dec Width	1-5	1-4	2	2
Mem Width	16b/cycle	8b/cycle	8b/cycle	4b/cycle
HDL	Chisel3	Chisel2	BSV+CMD	System Verilog

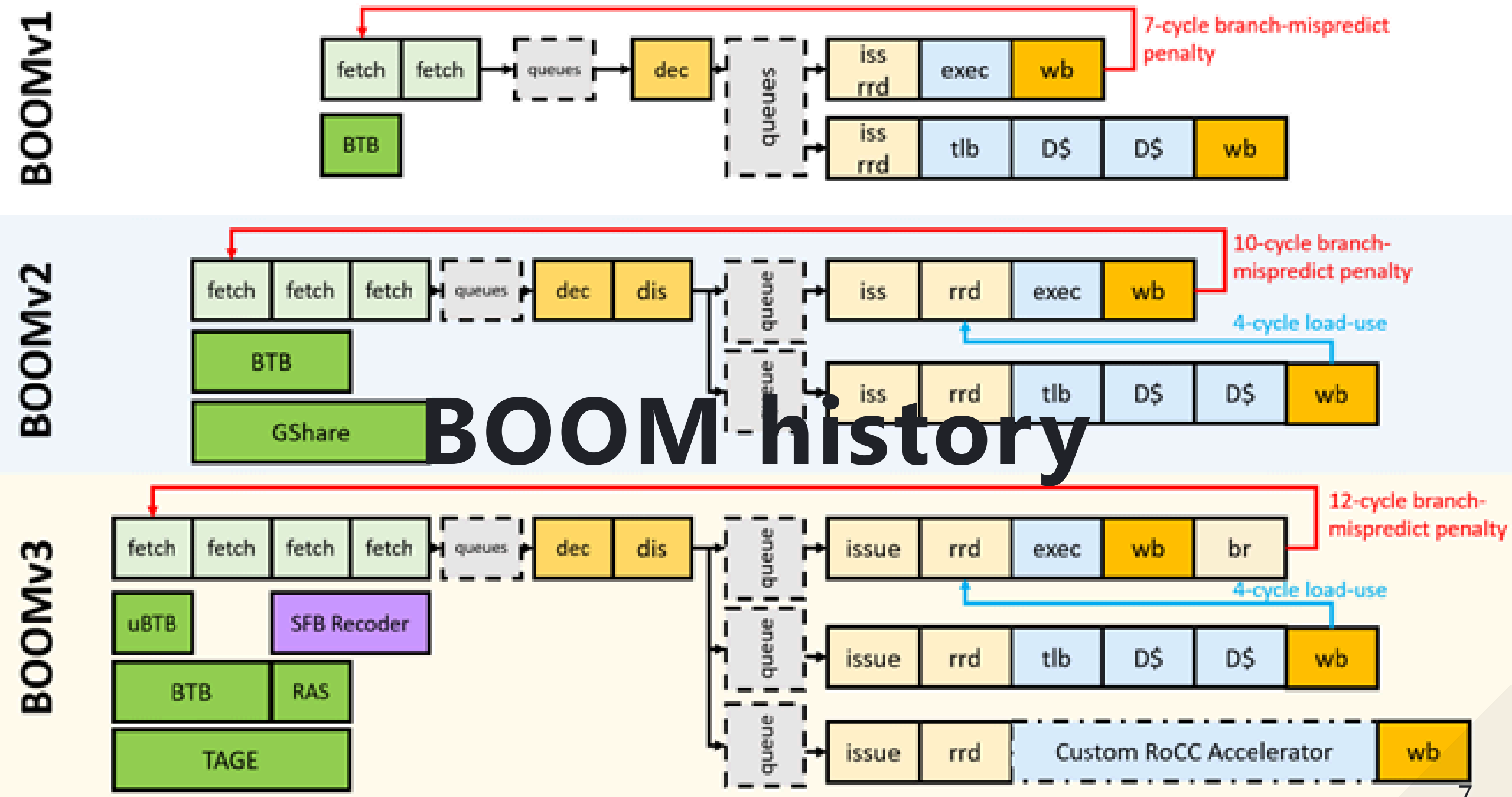


Figure 2: BOOM pipeline across BOOMv1, BOOMv2, and BOOMv3 (SonicBOOM)

- BOOM Version 1
 - Education tool for University
 - Based on microprocessor MIPS R10K (A RISC implementation of MIPS IV ISA)

